

IT222 — Database Principles

Dr. Soliman Fakeeh Hospital

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Today's Agenda

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Project Overview

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Work Distribution

What is this system?

Integrated Hospital Management

A centralized Database Management System(DBMS)
Designed to streamline Dr. Soliman Fakeeh Hospital's
Daily operations.

- ✓ Automates patient registration and medical history tracking.
- ✓ Coordinates real-time appointment scheduling
- ✓ Manages pharmacy inventory and medicine dispensing



View Description

This application is intended for hospital *administrative* and *medical staff* at Dr. Soliman Fakeeh Hospital.

Users of this view include:

- **Receptionists**
- **Nurses**
- **Doctors**

They are responsible for handling patient-related tasks and daily hospital operations.

5 Core Entities

Doctor

PK: DoctorID

Patient

PK: FileNumber

Appointment

PK: APP_ID

Department

PK: DeptCode

Medicine

PK: CommercialName

Entity Relationships

Department

1 : *

Doctor

one-to-many relationship

Department

1 : *

Appointment

one-to-many relationship

Patient

1 : *

Appointment

one-to-many relationship

Doctor

* : 1

Appointment

one-to-many relationship

Doctor

* : *

Medicine

many-to-many → junction table

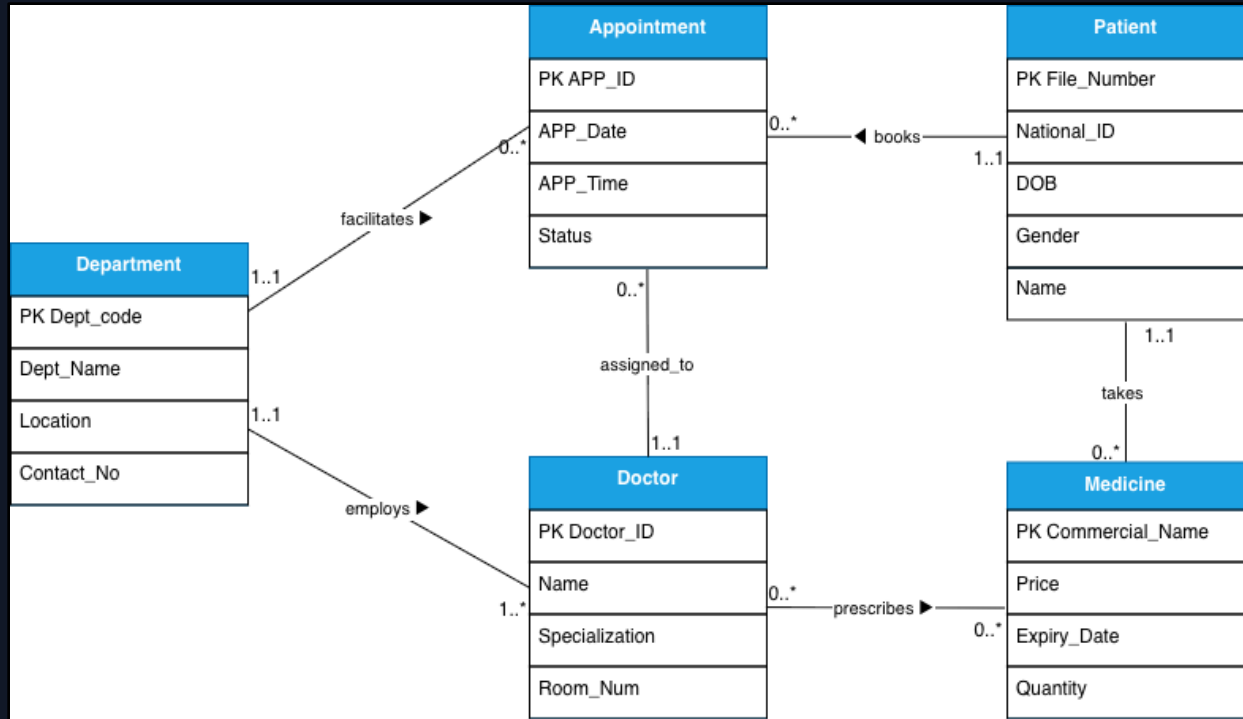
Patient

* : *

Medicine

many-to-many → junction table

Enhanced Entity-Relationship



Relational Schema

Doctor

DoctorID · Name · Specialization · RoomNum · Dept_code (FK)

Patient

FileNumber · NationalID · DOB · Gender

Department

Dept_code · Dept_Name · Location · ContactNo

Appointment

APP_ID · APP_Date · APP_Time · Status · DoctorID (FK) · FileNumber (FK) · Dept_code (FK)

Medicine

CommercialName · Price · Expiry_Date · Quantity

Prescribes

DoctorID (FK) · CommercialName (FK) – junction

Takes

FileNumber (FK) · CommercialName (FK) – junction

Insert into *Table Appointment*:

Insert appointment for a patient with file number 100.

The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Dana Alturaifi' are also visible. The main area is titled 'SQL Commands' and shows the schema 'WKSP_DBPROJECTSTUDEN'. The language is set to 'SQL' and the number of rows is set to '10'. The SQL command being executed is:

```
1 INSERT INTO Appointment (APP_ID, APP_DATE, APP_TIME, STATUS, DEPT_CODE, FILENUMBER, DOCTORID)
2 VALUES (1, TO_DATE('2026-05-10', 'YYYY-MM-DD'), 1000, 'Confirmed', 'D01', 100, 101);
```

Below the command, the 'Results' tab is selected, showing the output: '1 row(s) inserted.' and '0.05 seconds'. The footer contains the user email 'danalturaifi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeh_hosp', the language 'en', the copyright notice 'Copyright © 1999, 2024, Oracle and/or its affiliates.', and the version 'Oracle APEX 24.2.14'.

Find Doctors by Department

List all doctors assigned to the Cardiology department.

The screenshot shows the APEX SQL Workshop interface. At the top, there's a navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Dana Alturafi' are on the right. Below the navigation bar, the 'SQL Commands' section is active, showing a query in the 'Language' dropdown set to 'SQL'. The query is:

```
1 SELECT doc.NAME, doc.SPECIALIZATION, doc.ROOMNUM
2 FROM DOCTOR doc
3 JOIN DEPARTMENT dept ON doc.DEPT_CODE = dept.DEPT_CODE
4 WHERE dept.DEPT_NAME = 'Cardiology';
```

The 'Results' tab is selected, displaying a table with three columns: NAME, SPECIALIZATION, and ROOMNUM. The table contains three rows of data:

NAME	SPECIALIZATION	ROOMNUM
Dr. Nora	Orthopedic	102
Dr. Khalid	Cardiologist	101
Dr. Ahmed	General	103

Below the table, it states '3 rows returned in 0.03 seconds' with a 'Download' link. At the bottom, there's a footer with contact information, a copyright notice 'Copyright © 1999, 2024, Oracle and/or its affiliates.', and the version 'Oracle APEX 24.2.4'.

Check Medicine Stock

Checks the current quantity of Panadol in the pharmacy.

The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Dana Alturaifi' are on the right. The main area is titled 'SQL Commands' and shows a query in the 'SQL' language. The query is: `SELECT * FROM APPOINTMENT WHERE STATUS = 'Confirmed' AND APP_DATE = TO_DATE('05/15/2026', 'MM/DD/YYYY');`. Below the query editor, the 'Results' tab is active, displaying a table with 7 columns: APP_ID, APP_DATE, APP_TIME, STATUS, DEPT_CODE, FILENUMBER, and DOCTORID. The table contains one row of data. Below the table, it states '1 rows returned in 0.04 seconds' with a 'Download' link. The footer shows the user's email 'danalturaifi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeh_hosp', the copyright notice 'Copyright © 1999, 2024, Oracle and/or its affiliates.', and the version 'Oracle APEX 24.2.16'.

Language: SQL Rows: 10 Clear Command Find Tables Save Run

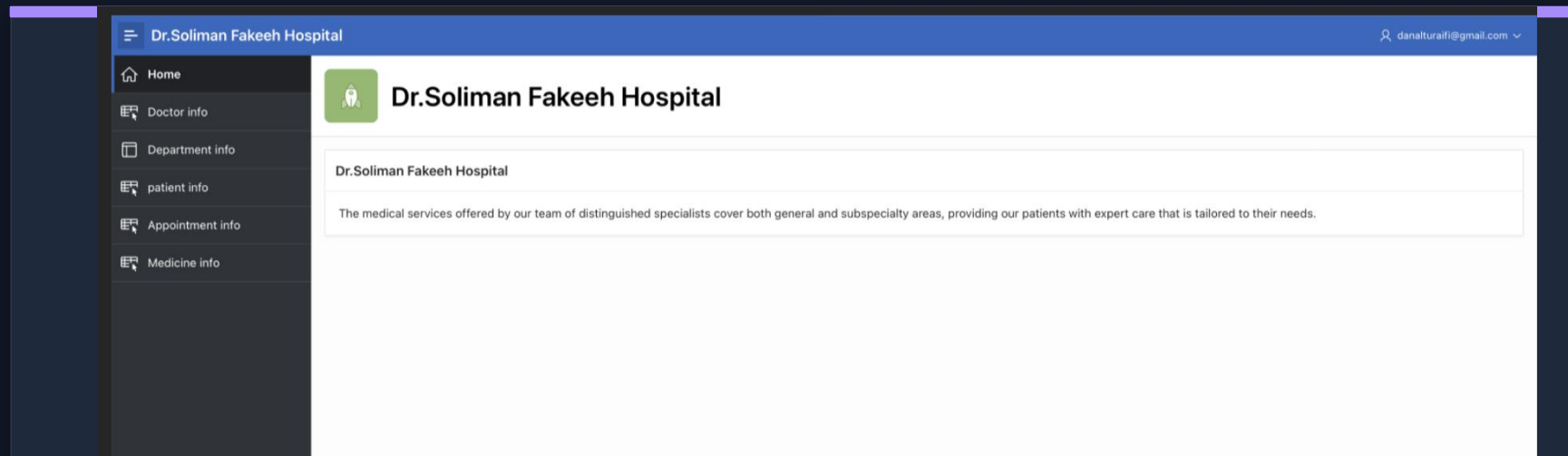
```
1 SELECT * FROM APPOINTMENT WHERE STATUS = 'Confirmed' AND APP_DATE = TO_DATE('05/15/2026', 'MM/DD/YYYY');
```

APP_ID	APP_DATE	APP_TIME	STATUS	DEPT_CODE	FILENUMBER	DOCTORID
2	5/15/2026	1100	Confirmed	D01	200	102

1 rows returned in 0.04 seconds Download

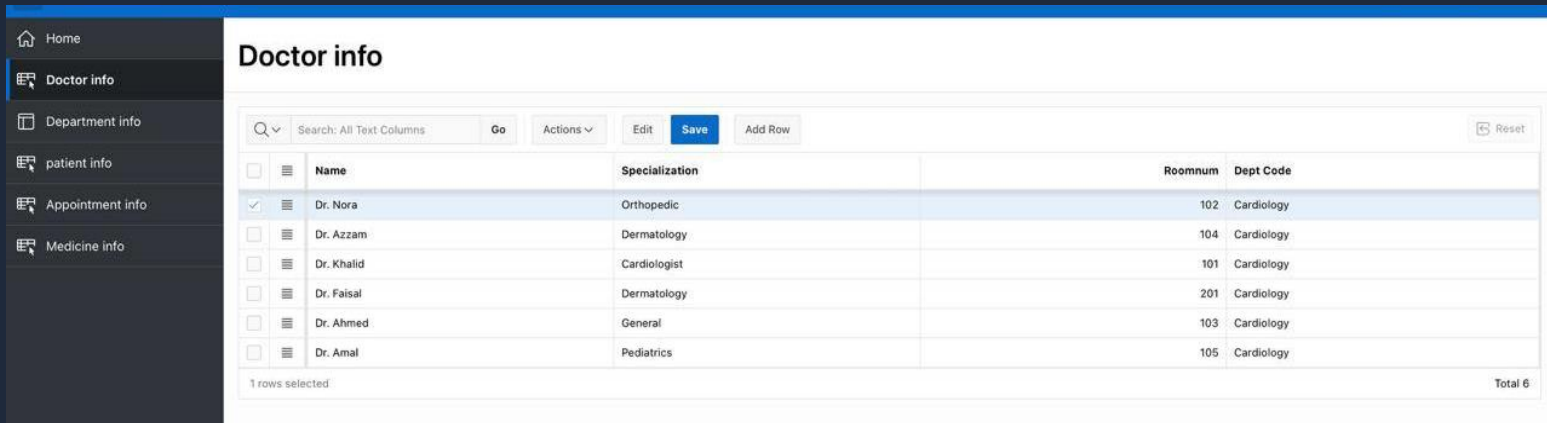
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Home Page



Main entry point with hospital overview and side navigation menu linking all sections.

Interactive Grid



The screenshot displays an Oracle APEX application interface. On the left is a sidebar with navigation links: Home, Doctor info (selected), Department info, patient info, Appointment info, and Medicine info. The main area is titled 'Doctor info' and contains an interactive grid. The grid has a search bar, action buttons (Edit, Save, Add Row), and a table with columns: Name, Specialization, Roomnum, and Dept Code. The table lists six doctors, with the first row (Dr. Nora) selected. A status bar at the bottom indicates '1 rows selected' and 'Total 6'.

	Name	Specialization	Roomnum	Dept Code
☒	Dr. Nora	Orthopedic	102	Cardiology
☐	Dr. Azzam	Dermatology	104	Cardiology
☐	Dr. Khalid	Cardiologist	101	Cardiology
☐	Dr. Faisal	Dermatology	201	Cardiology
☐	Dr. Ahmed	General	103	Cardiology
☐	Dr. Amal	Pediatrics	105	Cardiology

1 rows selected Total 6

Add**Edit****Delete**

Full CRUD on Doctor, Patient, Appointment, and Medicine records.

Master-Detail View

Dr.Soliman Fakeeh Hospital

danaturea@gmail.com

Home
Doctor info
Department info
patient info
Appointment info
Medicine info

Department info

Save

Search: All Text Columns Go Actions Edit Add Row Reset

☐	Dept Name	Location	Contactno
☐	Radiology	Building B	112224444
☒	Cardiology	Building A	112223333
☐	Laboratory	Building C	112225555
☐	Emergency	Building A	112226666

1 rows selected Total 4

Search: All Text Columns Go Actions Edit Add Row Reset

☐	Name	Specialization	Roomnum
☒	Dr. Nora	Orthopedic	102
☐	Dr. Azzam	Dermatology	104
☐	Dr. Khalid	Cardiologist	101
☐	Dr. Faisal	Dermatology	201
☐	Dr. Ahmed	General	103
☐	Dr. Amal	Pediatrics	106

1 rows selected Total 6

Department Info

Doctor Info

Select a department → doctors auto-filter

Work Distribution

Table Creation

Dana Alturaifi · 

20%

Data Insertion

Wajd Alotaibi · 

20%

Data Queries

Najla Almazyad · 

20%

App Builder

Ghala Alharbi · 

20%

App Builder

Sara Almazroa · 

20%

Thank You

Questions & Answers

Dr. Soliman Fakeeh Hospital · IT222 · Phase 3 · 1447H